Calculate gravitational force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate a gravitational interaction with consistent units.

SUCCESS CRITERIA - I can...

1. Identify both masses and the center separation.
2. Square the distance before dividing.
3. Check units and the scale of the result.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A ball is dropped from rest and falls ideally for 3 s. Use $g=10\ m/s^2$. Find its speed and direction.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $F = Gm_1m_2/r^2$; $G = 6.67\times10^{-11}\ N\cdot m^2/kg^2$. r is center-to-center separation. CPA may use the supplied numerator.

Worked reasoning: In a supplied gravity model, $Gm_1m_2=18\ N\cdot m^2$ and $r=3$ m. Square first: $r^2=9\ m^2$, so $F=18/9=2$ N, attractive.

Your turn: Sketch the two centers, label r and add the two force arrows. Show how the units reduce to newtons.



Marzano check - circle: ① ② ③

03 Read and interpret the model

For spheres 4 m apart center to center, a student substitutes r=2 m because each is 2 m from the midpoint. Identify the error.



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

04 Apply the physics

A gravity simulator supplies $Gm_1m_2 = 16\ N\cdot m^2$. Find the force at each center-to-center separation in the table. Describe what happens when the distance doubles.

Separation r (m)	r^2 (m^2)	Force (N)
4	____	____
8	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Why can a tiny force between classroom-sized objects be physically real even if you cannot feel it?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

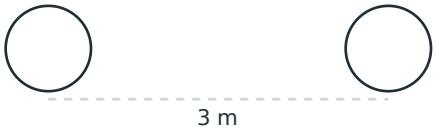
A 4 kg ball is moving upward at 6 m/s after leaving a hand. Ignore air resistance; use g=10 N/kg. Draw and justify its free-body diagram.
Draw the forces on the chosen object and label their sources.

Add and label force arrows.

Marzano check - circle: 1 2 3

07 Exit check

Given $Gm_1m_2=36\ N\cdot m^2$ and r=3 m, find force and direction.



Mass 1 Mass 2

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Keep the numerator 18 $N\cdot m^2$. Find the force at 6 m, then design a separation that gives 18 N.

Separation (m)	Force (N)
___	___
___	___

Marzano check - circle: 1 2 3